

## Inputs

|                                  |          |                              |
|----------------------------------|----------|------------------------------|
| Positive electrode thickness [m] | 7.56e-05 | parameter set                |
| Negative electrode thickness [m] | 8.52e-05 | parameter set                |
| Separator thickness [m]          | 1.2e-05  | parameter set                |
| Positive porosity                | 0.335    | parameter set                |
| Negative porosity                | 0.3      | override params/fcc.json     |
| Separator porosity               | 0.47     | parameter set                |
| Positive density [kg/m3]         | 3262     | parameter set                |
| Negative density [kg/m3]         | 1657     | parameter set                |
| Area [m2]                        | 0.1027   | parameter set (0.065 x 1.58) |
| Positive particle radius [m]     | 4e-06    | override                     |
| Negative particle radius [m]     | 2.5e-06  | override                     |
| Electrolyte sigma [S/m]          | 2        | override (estimate)          |
| Electrolyte D [m2/s]             | 2.5e-10  | override (estimate)          |
| t+                               | 0.55     | override (estimate)          |
| SEI kinetic rate constant [m/s]  | 2e-15    | override (estimate)          |
| Cooling h [W/m2K]                | 60       | override (design)            |

## CapacityEnergy

|                              |                      |                          |
|------------------------------|----------------------|--------------------------|
| 1C discharge capacity [Ah]   | 5.046876067471801    | cell/r2_fcc_1c_dfn.json  |
| Discharge energy [Wh]        | 18.17764516890772    | cell/r2_fcc_energy.json  |
| Mass (contract caliber) [kg] | 0.04272958670760001  | calc-energy              |
| ED [Wh/kg]                   | 425.4112096449225    | calc-energy: energy/mass |
| ED [Wh/L]                    | 881.4617464372169    | calc-energy              |
| Volume [m3]                  | 2.062216e-05         | calc-energy              |
| Thickness [m]                | 0.0002008            | calc-energy              |
| Midpoint voltage [V]         | 3.877180034738062    | calc-energy              |
| DCR [ohm]                    | 0.002891769175844239 | calc-energy              |
| Power density [W/kg]         | 33881.90921952434    | calc-energy              |

## NP\_Mass

|                             |                   |                                                                                                                      |
|-----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------|
| N/P ratio (density caliber) | 0.965854154102845 | $\rho \times (1 - \epsilon) \times L \times \text{usable stoich}$ ; neg 0.372 Ah/g x dx0.85, pos 0.274 Ah/g x dy0.72 |
| Qn areal [Ah/m2]            | 0.03              | formula                                                                                                              |
| Qp areal [Ah/m2]            | 0.03              | formula                                                                                                              |
| Layer masses [g]            | 42.73             | calc-energy layer_kg_m2 x area x 1000                                                                                |

## Process

|                                     |                                |                                                   |
|-------------------------------------|--------------------------------|---------------------------------------------------|
| Positive areal density [g/m2]       | 0.2                            | thickness x (1-porosity) x density                |
| Negative areal density [g/m2]       | 0.1                            | formula                                           |
| Positive compaction density [g/cm3] | 2.17                           | density x (1-porosity)/1000                       |
| Negative compaction density [g/cm3] | 1.16                           | formula                                           |
| Electrolyte fill [g]                | 6.97                           | pore volume x 1.2 g/cm3 (lit. density, annotated) |
| Formation                           | 0.1C CC to 4.2V, 25C, 2 cycles | design recommendation; production tuning required |